GPC

Gen² 2026 | ICLR

An expressive and tractable deep generative model for genetic variation data

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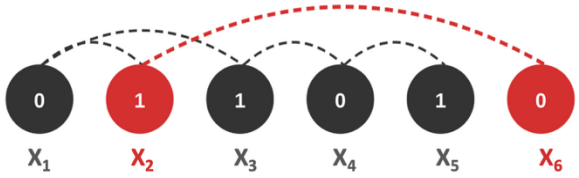
What is genetic variation data?

Variants (SNPs)

Individuals	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
ID_1	0	1	0	1	1	0	1	0
ID_2	1	1	0	0	1	1	0	1
ID_3	0	0	1	1	0	1	0	0
ID_4	1	0	1	0	0	1	1	0
ID_5	0	1	0	1	1	0	0	1
ID_6	1	0	1	0	1	0	1	0

Three key challenges

Complex Correlation

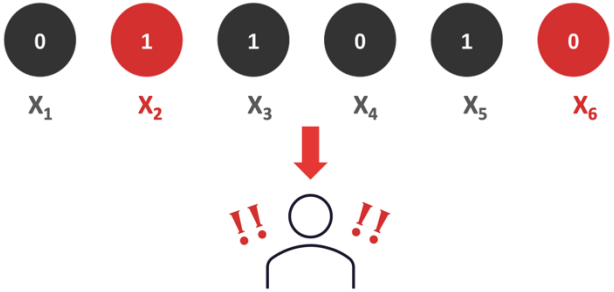


SNPs are correlated across different length scales

Imputation

Many SNPs are unobserved; need to infer missing genotypes

Privacy

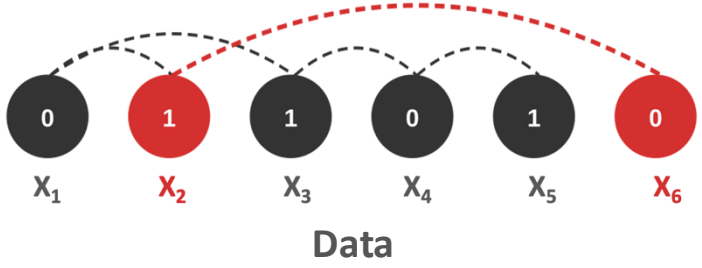


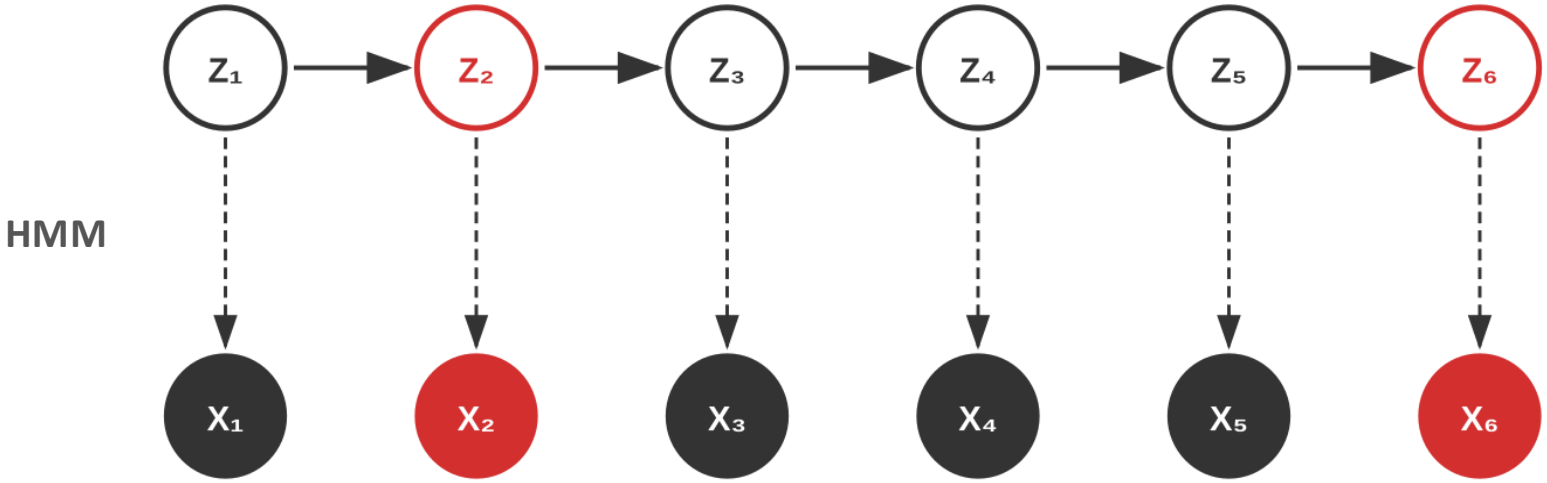
Real genomes can be linked back to individuals

A good model should capture correlation structure, enable imputation, and preserve privacy

Current models struggle to achieve these while also being **tractable**

Hidden Markov Models

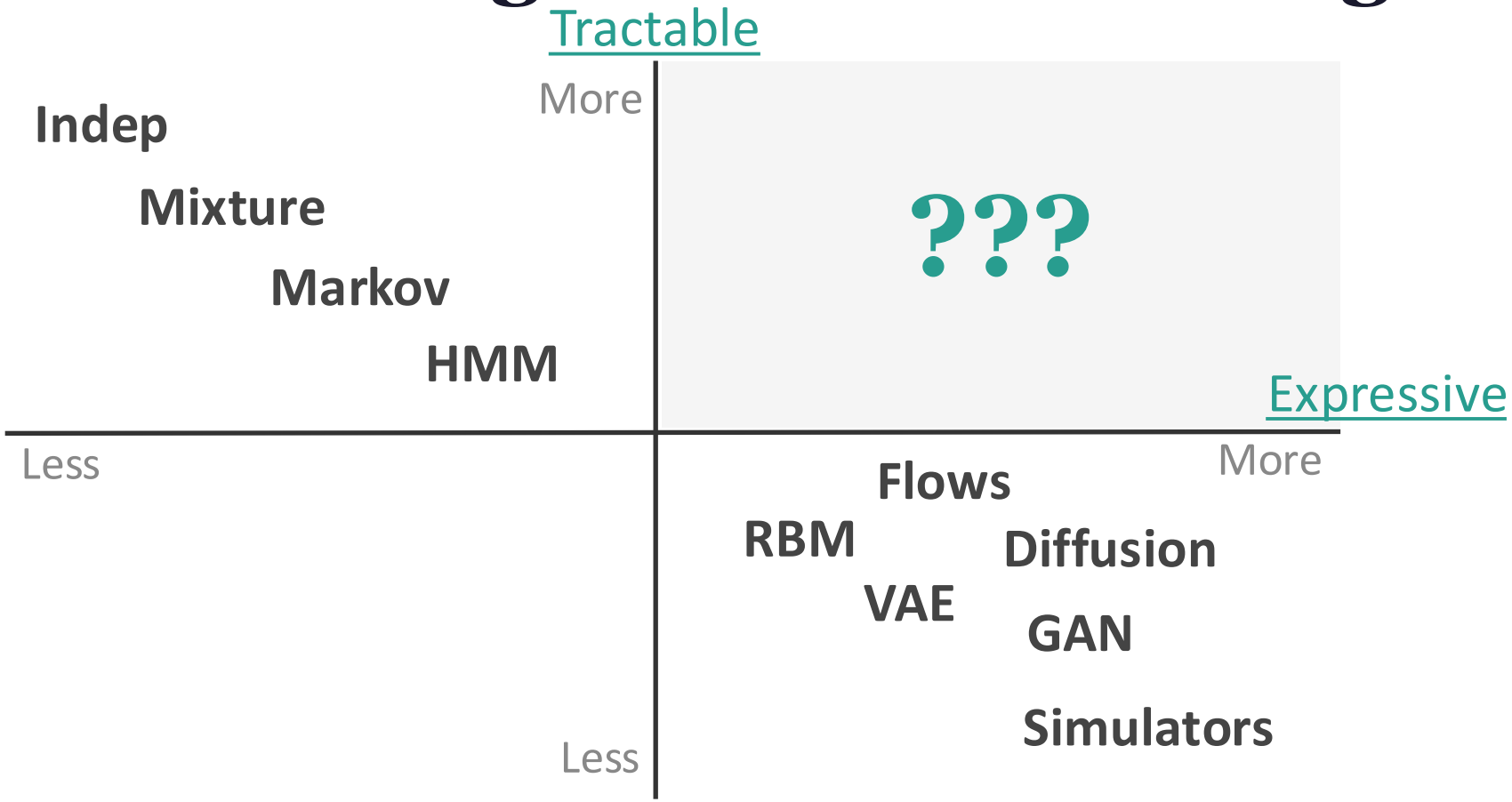




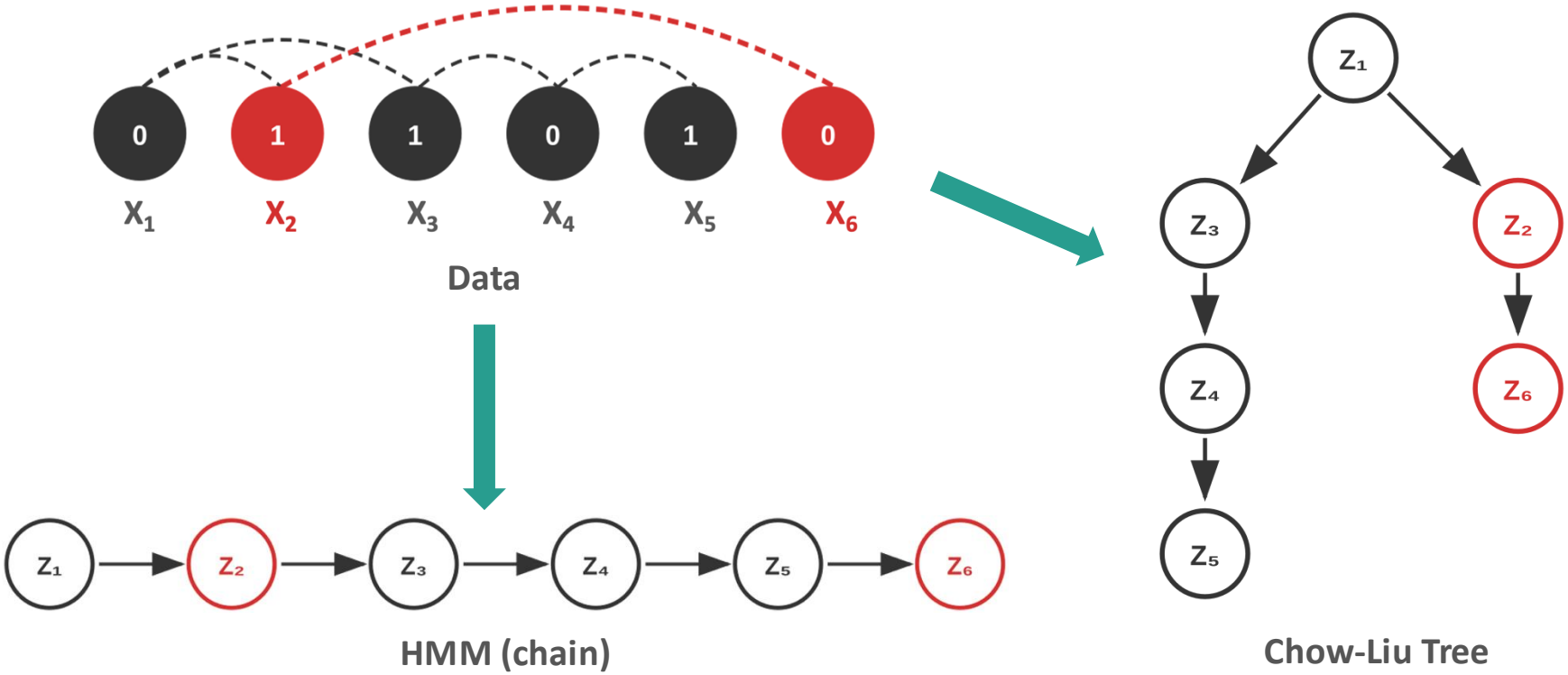
Not explicitly modeled

Chain structure forces information between distant SNPs through all intermediates

Tradeoffs in generative modeling



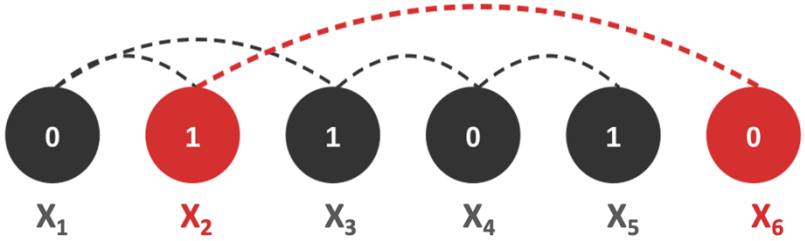
From chains to trees



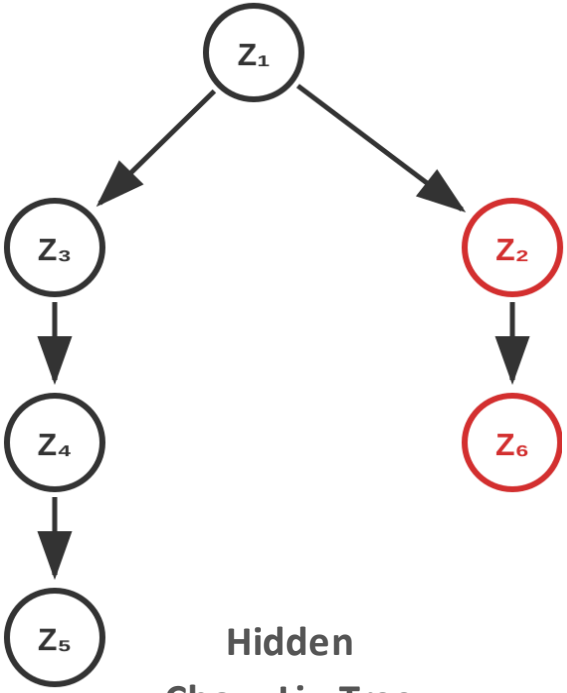
Tree structure places correlated SNPs close together regardless of genomic position

Chow and Liu IEEE 1968

Genetic Probabilistic Circuits (GPC)



Data



Hidden Chow-Liu Tree

Compile HCLT → **Probabilistic Circuit** → exact likelihoods, efficient training and inference

Vergari et al. AAAI 2020; Liu et al. ICML 2024; for detailed PC specification

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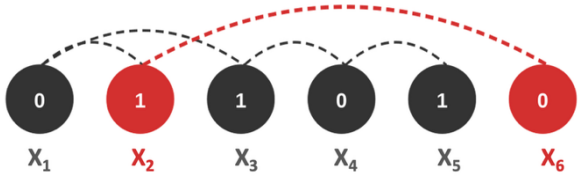

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Linkage Disequilibrium (LD)

Imputation

Many SNPs are unobserved; need to infer missing genotypes

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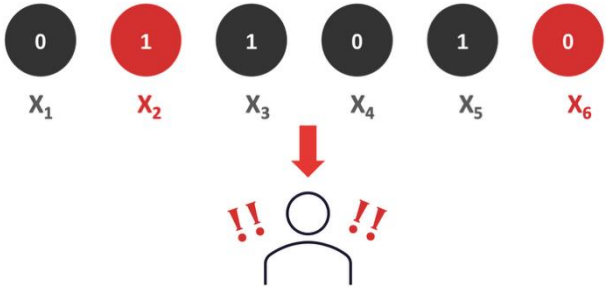
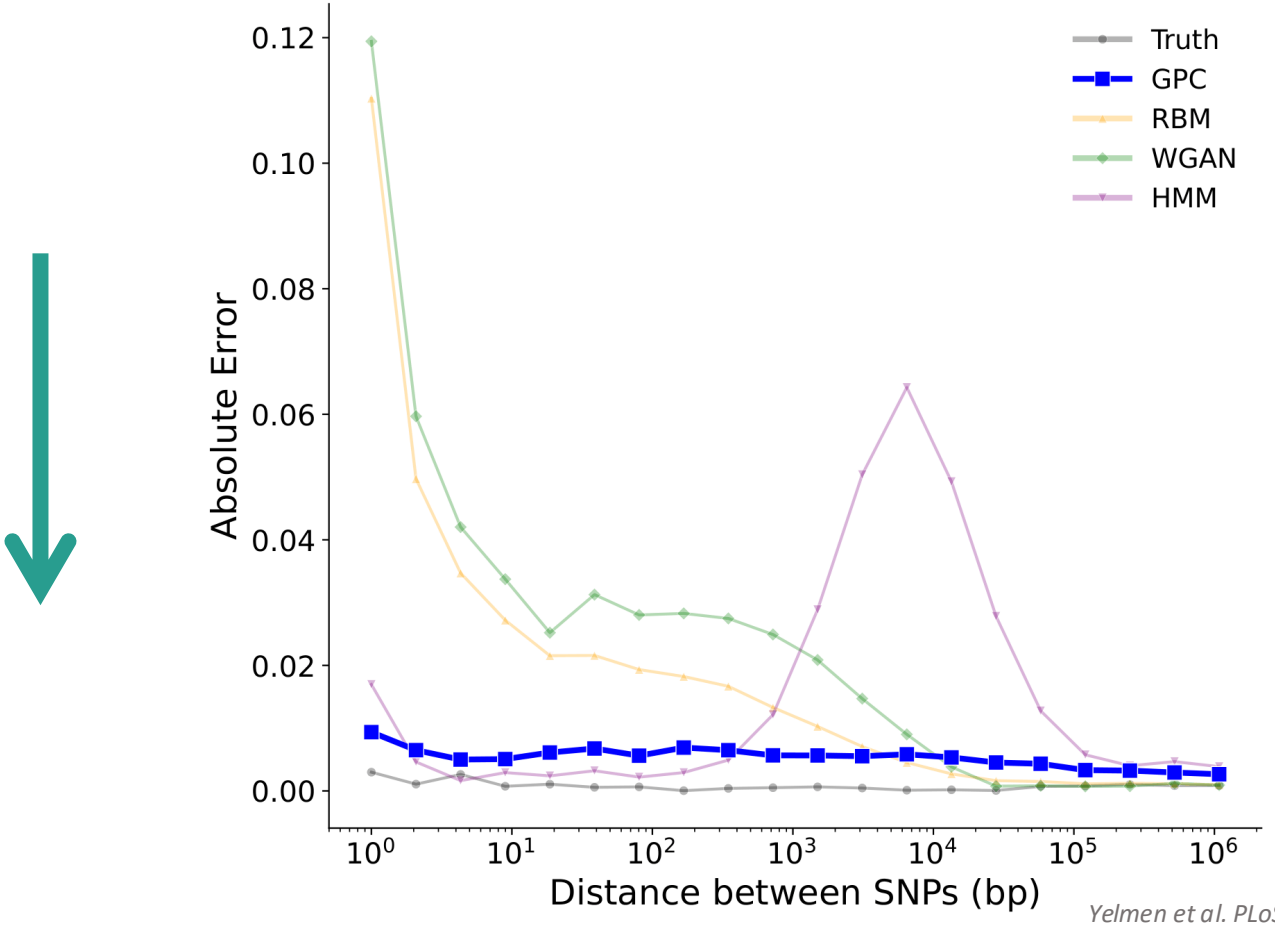


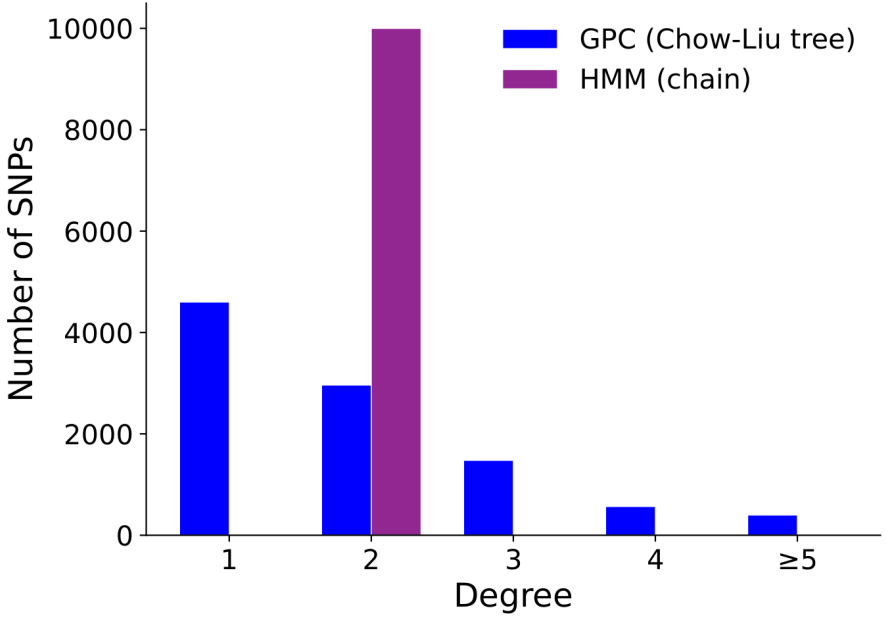
Real genomes can be linked back to individuals

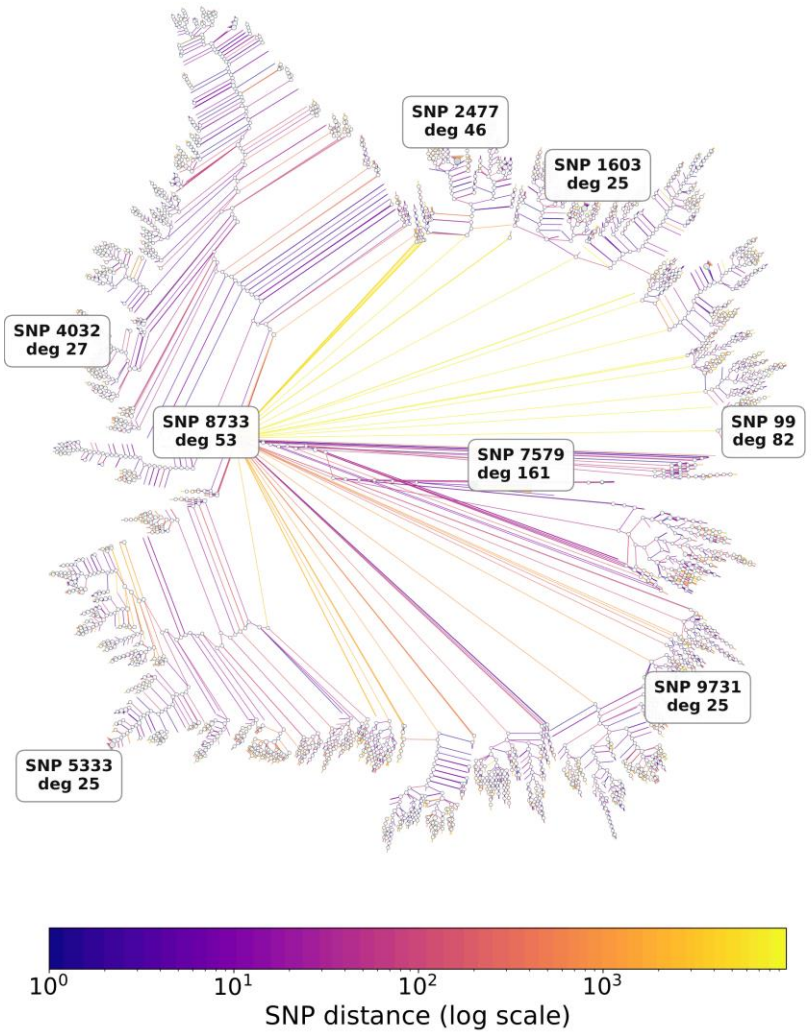
GPC closely reproduces LD patterns



Yelmen et al. PLoS Comp Bio 2023 (plotting scripts adapted)

GPC stores LD information more effectively





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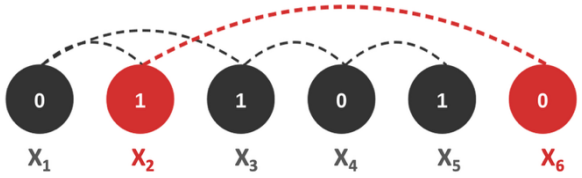

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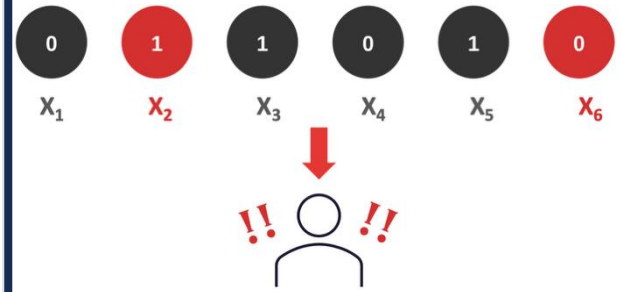
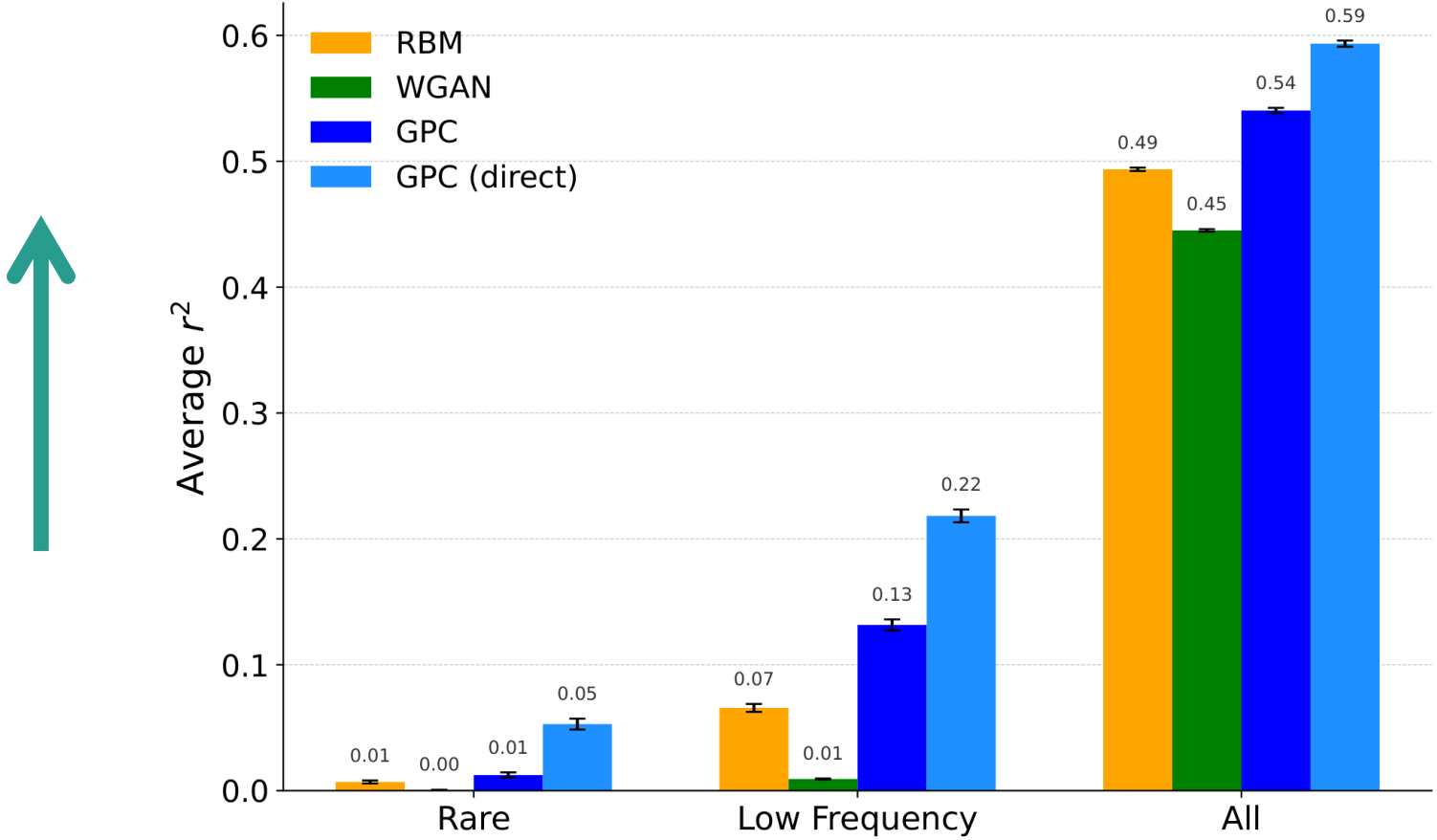


Real genomes can be linked back to individuals

GPC achieves best imputation among generative baselines



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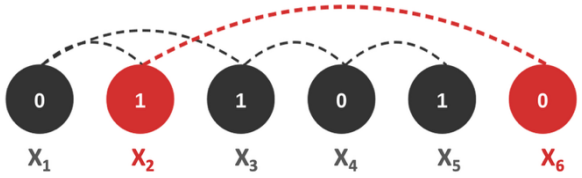

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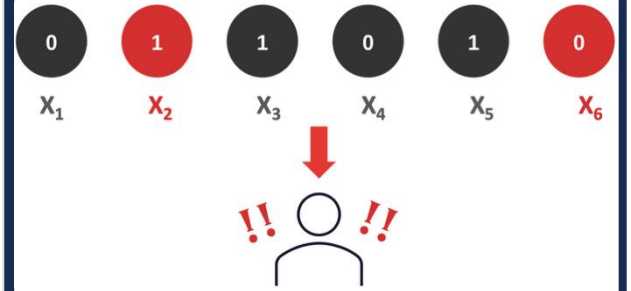
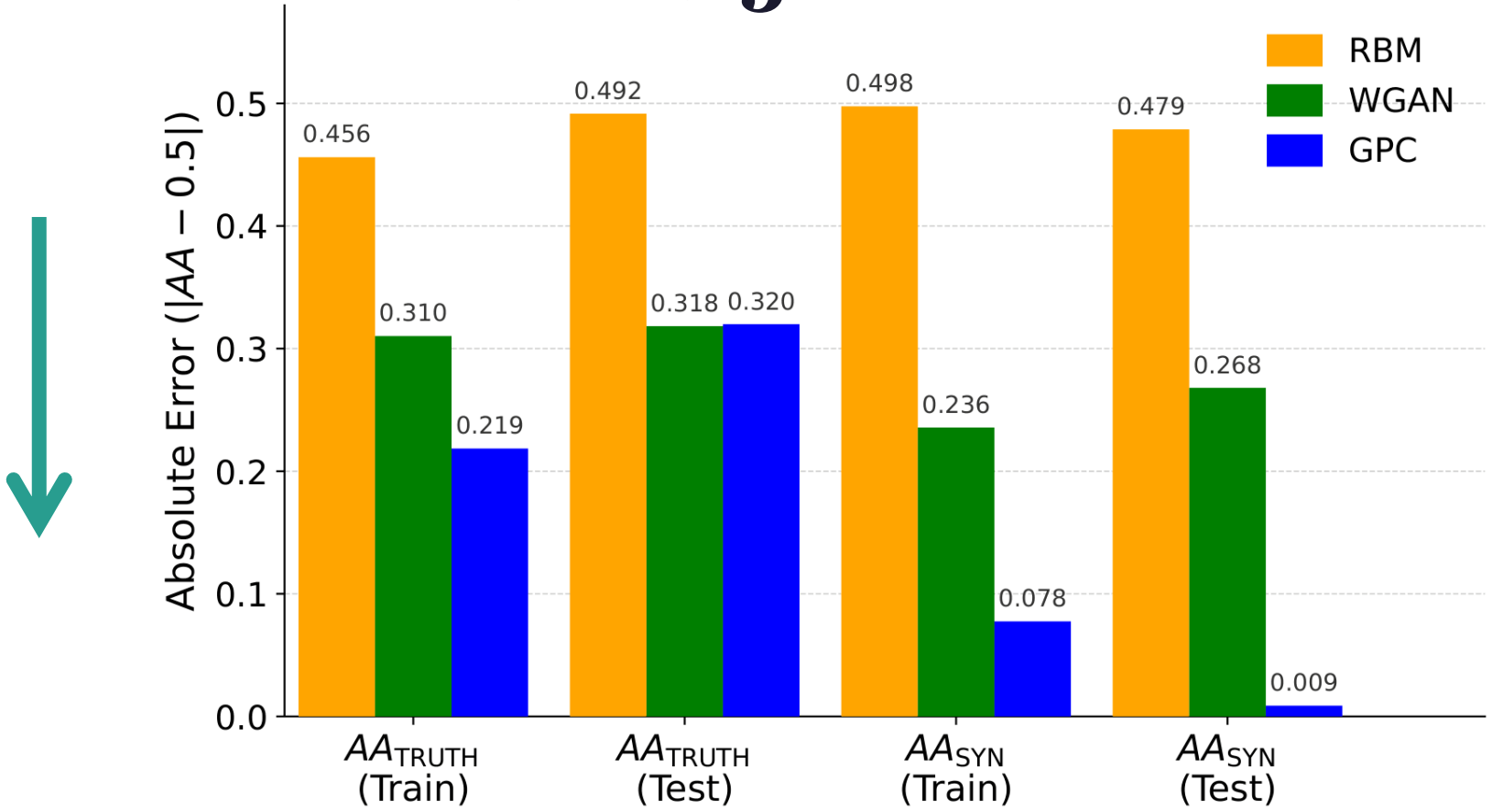


Real genomes can be linked back to individuals

Adversarial Accuracy (AA)
Based on nearest neighbors

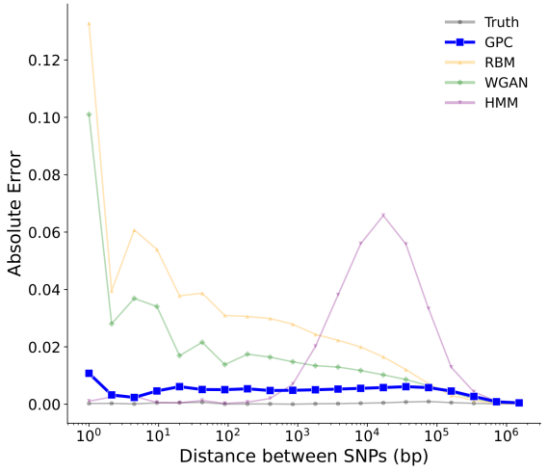
Yale et al. 2019

GPC balances privacy and utility: closest to ideal values of 0.5



All findings replicate in UK Biobank

Complex Correlation



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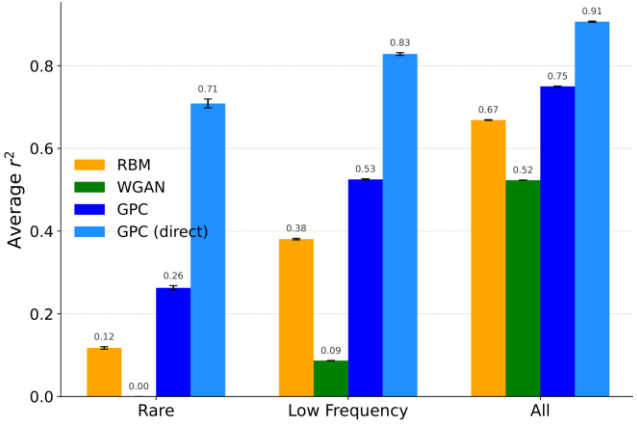

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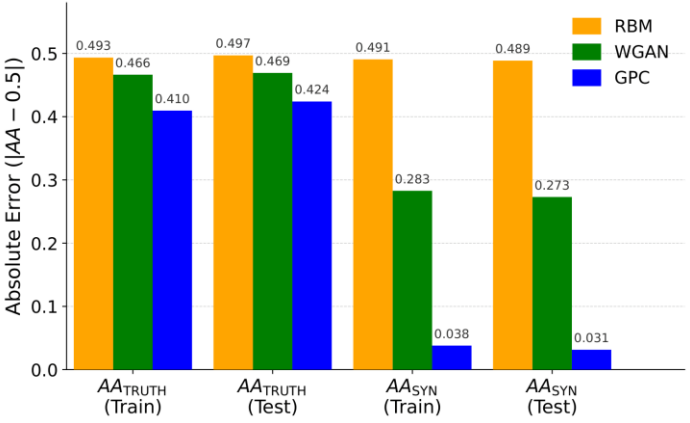

GPC Takeaways

1. Expressive, tractable, and efficient: tree structure captures LD at all scales with exact inference

2. Best imputation accuracy among generative models

3. Better privacy-utility tradeoff for synthetic data than existing methods

Acknowledgments

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Sriram Lab
UCLA

* Joint supervision

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Thank you!

github.com/sriramlab/GPC

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